SVM Constraint Transformation to Standard QP Form (Gx ≤ h, Ax = b)

Minimize: ½ $\alpha^n P \alpha + q^n \alpha$ Subject to: $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$ $0 \le \alpha_1, \alpha_2, \alpha_3, \alpha_4 \le 1$

$\alpha_1 \ge 0$, $\alpha_1 \le 1 \Rightarrow -\alpha_1 \le 0$, $\alpha_1 \le 1$

$(\forall i = 1..4)$

$$G = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$h = [\,0\ 0\ 0\ 0\ 1\ 1\ 1\ 1\,]^n$

Solution

$A = [\,1\ \ 1\ -1\ -1\,]$

$b = [\,0\,]$

Solution

$\alpha_1 = 0.0690$
$\alpha_2 = 0.0690$
$\alpha_3 = 0.0690$
$\alpha_4 = 0.000008$